

Exercise and worked solution

Futures & forwards

Four parts to handle forward contracts : lock a price with a short hedge, price a forward through cost of carry, tune a hedge ratio and a beta, then read the price curve and the convenience yield. Work through each part before reading the solution.

Part A · Short hedge and basis

PROBLEM

An oil producer will receive 2M bbl in three months. Spot is 59.50 and the three month futures $F_0 = 60$. It shorts 2,000 contracts of 1,000 bbl. Compute the net proceeds if spot ends at (a) 50, (b) 68. What exactly does the hedge lock in ?

SOLUTION

1. (a) At 50 : physical sale = $2M \times 50 = 100M$; futures gain = $2M \times (60 - 50) = 20M$; net = **120M**.
2. (b) At 68 : physical sale = $2M \times 68 = 136M$; futures loss = $2M \times (60 - 68) = -16M$; net = **120M**.
3. In both cases net proceeds are 120M, exactly 60 USD/bbl : the hedge locks in the initial futures price $F_0 = 60$, not the initial spot 59.50. The only residual uncertainty is the basis (spot minus futures at the close of the position).

Part B · Cost of carry, without and with income

PROBLEM

An asset trades at $S = 70$, the one year rate is $R = 5\%$. (a) One year forward price if it pays no income ? (b) And if it pays a coupon of 5 in 6 months (6 month rate $R_6 = 4\%$) ?

SOLUTION

1. (a) No income : $F = S(1+R)^T = 70 \times 1.05 = \mathbf{73.50}$. The forward exceeds spot by the financing cost of the position alone.
2. (b) With income, I is the present value of the coupon : $I = 5/1.04^{0.5} = 5/1.0198 = 4.903$. Then $F = (S - I)(1+R)^T = (70 - 4.903) \times 1.05 = 65.097 \times 1.05 = \mathbf{68.35}$.
3. Income lowers the forward : the holder collects the coupon, so they finance only S less the present value of that income. Classic trap : forgetting to discount the coupon before subtracting it.

Part C · Optimal hedge ratio and steering beta

PROBLEM

(a) An airline hedges 1M gallons of jet fuel with heating oil futures (contracts of 42,000 gallons) ; $\rho = 0.9$, $\sigma_S = 0.025$, $\sigma_F = 0.03$. Optimal ratio h^* , effectiveness, number of contracts ? (b) A 1M portfolio has $\beta = 1.25$; the S&P 500 mini futures is worth $50 \times$ index, index at 2,500. Contracts for a full hedge, then to bring β to 0.5 ?

SOLUTION

1. (a) $h^* = \rho \sigma_S / \sigma_F = 0.9 \times 0.025 / 0.03 = \mathbf{0.75}$: only 75% of the exposure is hedged, since jet fuel and heating oil do not move perfectly together. Effectiveness = $\rho^2 = 0.81$ (81% of variance hedged), note it is ρ^2 not ρ . Contracts : $N^* = h^* \times Q_A / Q_F = 0.75 \times 1,000,000 / 42,000 = 17.86 \approx \mathbf{18 \text{ contracts}}$.
2. (b) Contract value $V_F = 50 \times 2,500 = 125,000$. Full hedge : $N^* = \beta \times V_A / V_F = 1.25 \times 1,000,000 / 125,000 = \mathbf{10 \text{ contracts short}}$; the hedged portfolio then earns the risk free rate.
3. To steer β toward 0.5 take the beta difference : $N = (\beta - \beta^*) \times V_A / V_F = (1.25 - 0.5) \times 8 = \mathbf{6 \text{ contracts short}}$ (with $V_A / V_F = 1,000,000 / 125,000 = 8$). Trap : using β alone would bring it to 0, not to the target.

Part D · Price curve, arbitrage and convenience yield

PROBLEM

- (a) An index trades at 2,500, $R = 5\%$, dividend yield $Q = 3\%$. 6 month forward ? The market quotes 2,540 : which arbitrage ?
- (b) For oil $S = 69$, 6 month futures $F_6 = 65$, storage cost $U = 1$ (in PV), $R = 2\%$. Is the forward below spot an arbitrage ? Compute the convenience yield and interpret.

SOLUTION

1. (a) An index behaves like a yield paying asset : $F = S \times ((1+R)/(1+Q))^T = 2,500 \times (1.05/1.03)^{0.5} = 2,500 \times 1.00964 \approx \mathbf{2,524}$. The market forward (2,540) is too expensive, so cash and carry : borrow, buy the basket at spot (collecting dividends), sell the forward at 2,540 ; at maturity deliver, repay financing net of dividends ($\approx 2,524$) and pocket 16 risk free points. This selling pressure pushes the forward back toward 2,524.
2. (b) No arbitrage. Oil is a consumption asset : the reverse arbitrage assumes holders sell their stock to buy it back forward, which they refuse (they hold it to consume it). Only the upper bound $F \leq (S+U)(1+R)^T$ holds, and the price can sit well below it.
3. Convenience yield : solve $65 = 70 \times (1.02/(1+Y))^{0.5}$, so $(1+Y) = 1.02 \times (70/65)^2 = 1.183$, i.e. $Y \approx \mathbf{18.3\%}$. This high yield measures the value of having oil physically on hand : it rises when inventories are seen as low (fear of shortage), which justifies holding stock despite a positive carry cost and a futures below spot.

Key takeaways : a hedge locks in the initial futures price F_0 , not spot, leaving only the basis at risk ; cost of carry gives $F = S(1+R)^T$, adjusted for income ($F = (S-I)(1+R)^T$) or a yield ($F = S((1+R)/(1+Q))^T$) ; the optimal ratio is $h^* = \rho \cdot \sigma_S / \sigma_F$ and effectiveness is ρ^2 ; beta is steered by $(\beta - \beta^*) \cdot V_A / V_F$; and for a consumption asset only the upper bound holds, the gap read as a convenience yield.